

AVNI • result report

generated 2026-09-20 21:04:21 UTC • AVNI-VL 0.9 • SIH 26167 • SAC/ISRO

area of interest	MG Road 0.3 km NW • Bengaluru, Karnataka
AOI centroid	12.9744°N 77.6093°E
query raised	2026-09-20 21:04:15 UTC
session	AVNI-VL 0.9 • SIH 26167 • SAC/ISRO

Imagery

primary scene	S2_L2A_2025-08-14_tile43REP.jp2
Sentinel-2 L2A • S2 L2A • tile 43REP • acquired 2025-08-14 05:17:42Z	
SAR scene	RISAT-1A_GRD_250813.tif
RISAT-1A (C-band) • RISAT-1A • GRD • asc • acquired 2025-08-13 00:41:09Z	

Question & answer

Where is flooding most severe, and can you trust the water reading near the buildings?

Flooding is most severe on the river terraces just south of the corridor spine — 4.7 ha of standing water at NDWI 0.61, with the deepest signature against the retaining walls of the older built-up rows. Near the buildings the reading is honest but incomplete: SAR backscatter of -3.2 dB is brighter than open water should be, which is the classic double-bounce inversion of floodwater against vertical structure. Act on the open-terrace water confidently; treat any pixel within ~50 m of a structure as unverified until the geometry check clears.

Confidence & Verification

consistency score	0.47
gate	passed – at or above 0.45
reason	Terrace water holds, but the built-up edge splits the classifier. Stable across 4 of 8 image orientations.

Physics check

NDWI	0.61 (water gate 0.45)
SAR backscatter	-3.2 dB
verdict	possible urban flood double-bounce inversion – SAR brightness near building edges suggests floodwater interacting with structures, not dry land
flag	flagged – readings disagree

answer bank NDWI / sigma0 readings • scale and marker positions derived

Geodetic Evidence

CRS	EPSG:4326
AOI area	3684.1 ha (36.84 km ²)
AOI vertices	4
footprint area (fixture)	4.7 ha

UTM 43N · 783074 E 1435754 N
footprint width 22.03 km x 12.30 km
ground sample distance 16 m/px
georeference declared scene footprint

WGS84 radii of curvature at the ring centroid, shoelace on the tangent plane



map snapshot · EPSG:4326 · full scene · scale bar 500 m · AOI outline dashed, 4 vertices

Execution Trace

1. NDWI computed
geodesy.py
0.61
2. SAR backscatter computed
physics_check.py
-3.2 dB
3. Consistency check run
abstain.py
0.47 (4/8 stable)
4. Answer generated
vision-language model

4 steps · sources and values as shown in the execution trace panel

Geometry cross-reference: avni_answer_where-is-flooding-most-severe-and-can-yo.geojson